

Volatility forecasting with ECB news stance

An empirical investigation of stance-augmented GARCH models

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Abstract

This paper investigates whether European Central Bank communication contains incremental information for forecasting Euro Stoxx 50 volatility beyond standard GARCH-family models. We construct daily hawkish and dovish stance indices from 1,159 ECB documents published between 2007 and 2023, including Monetary Policy Statements, Monthly and Economic Bulletins, and speeches by the ECB President and Chief Economist. Each document is segmented into sentences and scored with FOMC-RoBERTa, producing sentence-level hawkish, dovish, and neutral probabilities that are aggregated to daily stance series. These inputs are incorporated into News-Augmented GARCH specifications following Sadik, Date and Mitra (2018) and compared against GARCH, GJR-GARCH, and EGARCH benchmarks. Models are estimated on 2008–2018 data and evaluated out-of-sample over 2019–2023 using QLIKE, RMSE, Diebold–Mariano tests, Model Confidence Sets, forecast combinations, sub-sample analysis and robustness checks. The results provide no evidence that ECB stance improves volatility forecasts. GJR-GARCH and EGARCH consistently dominate the news-augmented specifications, while the optimal forecast-combination weight on the stance-augmented model is zero. The null result is robust to alternative communication sources, classifier thresholds, volatility proxies, innovation distributions and surprise-based stance constructions.

Keywords: GARCH, news stance, monetary policy communication, volatility forecasting, central-bank communication, FOMC-RoBERTa.

1 Introduction

Central banks do not move financial markets only through interest-rate decisions; the language of press conferences, policy statements and official speeches is itself a market-moving instrument. Whether ECB communication carries information about future market volatility, over and above what standard volatility models already extract from past returns, is an open empirical question. This paper addresses it directly. We construct a daily stance index from three sources of publicly available ECB communication (Monetary Policy Statements, Economic Bulletins, and speeches) over April 2007 to December 2023. Sentences are classified using FOMC-RoBERTa (Shah et al., 2023), a transformer that labels central-bank language as hawkish, dovish, or neutral. The resulting stance scores enter the News-Augmented GARCH framework of Sadik et al. (2018), with separate sensitivities for hawkish and dovish communication following Smales (2016). The target variable is the daily return volatility of the Euro Stoxx 50, evaluated out-of-sample against GARCH (Bollerslev, 1986) and GJR-GARCH (Glosten et al., 1993) benchmarks across two horizons and two estimation-window schemes. The results provide no evidence that ECB stance improves out-of-sample volatility forecasts beyond the asymmetric GARCH benchmarks. GJR-GARCH dominates the news-augmented specifications in every (scheme, horizon) cell, and the Model Confidence Set of Hansen et al. (2011) retains only the asymmetric benchmarks. The null is robust across source decompositions, confidence thresholds, innovation distributions, and three alternative surprise-encoding constructions.

Section 2 reviews the related literature. Section 3 describes the data and stance construction. Section 4 presents the econometric methodology. Section 5 reports the empirical results. Section 6 discusses limitations and future work, and Section 7 concludes.

2 Literature review

2.1 Volatility modeling and the GARCH framework

Modern empirical volatility modeling originates with the ARCH framework introduced by Engle (1982), which formalized the tendency of financial return volatility to cluster over time. Bollerslev (1986) subsequently generalized this approach to the GARCH(1,1) specification by incorporating lagged conditional variance dynamics, creating a parsimonious model that remains the standard benchmark in volatility forecasting applications. A key limitation of the standard GARCH specification is its inability to capture asymmetric volatility responses. Empirical evidence consistently shows that negative return shocks generate higher future volatility than positive shocks of the same magnitude. Glosten et al. (1993) addressed this issue through the GJR-GARCH specification, while Nelson (1991) proposed the EGARCH model, which captures asymmetry through a logarithmic variance formulation.

2.2 News sentiment and volatility forecasting

The increasing availability of machine-readable text data has generated a large literature examining whether news sentiment contains incremental information for financial markets and volatility forecasting. Among the most relevant contributions for the present study is Smales (2016), who examines the relationship between news sentiment, implied volatility, and stock returns in a time-varying framework. Using Thomson Reuters news scores, Smales documents that negative news exerts a substantially larger effect on implied volatility than positive news and that this relationship strengthens during periods of elevated market stress. These findings motivate the separate modeling of positive and negative news effects and suggest that sentiment volatility relationships may vary across market conditions. The primary methodological reference for the present study is the News-Augmented GARCH (NA-GARCH) framework proposed by Sadik et al. (2018). Their model incorporates textual news sentiment directly into the conditional variance equation through a multiplicative scaling mechanism, allowing sentiment to amplify or dampen the persistence of the underlying volatility. Using RavenPack affective sentiment scores for US equities, Sadik et al. (2018) show that the NA-GARCH specifications outperform the standard GARCH and EGARCH benchmarks in-sample and out-of-sample. More recently, Bodilsen and Lunde (2024) show that macroeconomic news sentiment significantly improves volatility forecasts, while firm-specific news contributes comparatively little predictive information. All of the aforementioned findings pertain to U.S. news measures and stock markets. The present study extends the research to the European equity market and focuses on applying the NA-GARCH framework by Sadik et al. (2018) to macroeconomic news, particularly ECB communication, inspired by the most recent finding by Bodilsen and Lunde (2024).

2.3 Central Bank communication and financial markets

A natural candidate for macroeconomic news in the euro area is ECB communication. Gürkaynak et al. (2005) show that the future-policy-path component of central-bank communication generates significant movements in asset prices beyond the policy-rate decision itself, and Cieslak and Schrimpf (2019) show that ECB press conferences and communication on the economic outlook produce equity and bond movements often comparable in size to conventional monetary-policy surprises. These findings motivate the use of ECB communication as the news input in our NA-GARCH framework. Within this literature, several studies have moved from generic affective sentiment toward hawkish/dovish stance classification, which is closer in spirit to the policy signal itself. Tobback et al. (2017) show that stance scoring of ECB press conferences tracks the policy cycle and improves Taylor-rule fits of the bank's reaction function. Shah et al. (2023) develop FOMC-RoBERTa, a transformer fine-tuned on hawkish/dovish/neutral labels for Federal Open Market Committee (FOMC) documents, and show that the resulting stance moves yields, equity prices, and macro indicators in theoretically consistent directions. Bernoth (2025) extends the Tobback et al. (2017) approach by constructing an ECB Communication Stance Indicator based on FOMC-RoBERTa extended by

classifications of monetary policy statements and finds that identified communication shocks contain forward-looking information distinct from conventional monetary-policy shocks. We follow this line by adopting stance rather than affective sentiment, but ask the distinct question of whether it improves out-of-sample volatility *forecasts* for euro-area equities. An important caveat, however, concerns the anticipation of ECB communication. Unlike unscheduled firm-specific news, ECB releases are highly anticipated through forward guidance, scheduled monetary-policy meetings, and continuous monitoring by professional analysts. Consequently, a substantial share of the informational content of ECB communication may already be incorporated into financial markets before official announcements occur, reducing the surprise component contained in stance indicators. In general, the existing literature suggests that macroeconomic news and central bank communication influence financial markets, but whether such information systematically improves volatility forecasting performance beyond standard GARCH dynamics remains an open empirical question.

3 Data

Our data set comprises three time series: daily equity log returns of the EuroStoxx 50 index and two directional daily macroeconomic news scores (derivation described in Section 3.2). Equity returns are only recorded on financial trading days; therefore, the news scores are aligned to the same trading-day frequency. The sample contains 4,201 days over the period April 2007 to December 2023 and is partitioned into three windows: i) pre-sample (02.04.2007 – 31.12.2007; 187 trading days) for burn-in but excluded from estimation, ii) in-sample (03.01.2008 – 28.12.2018; 2,754 trading days) for estimation, and iii) out-of-sample (03.01.2019 – 29.12.2023; 1,260 trading days) for forecasting evaluation. The split is fixed ex ante and never revisited. Log-returns exhibit pronounced negative skewness and excess kurtosis, particularly during the OOS window which captures the COVID shock (kurtosis 13.1), motivating the Student- t innovation assumption adopted throughout. The stance inputs are mildly positive on average ($\bar{S}_t > 0$) in-sample, reflecting a slight dovish tilt over the 2008–2018 period. See Appendix A, Figure 4, for a time-series plot.

Table 1: Descriptive statistics of daily log-returns and stance series.

Sample	T	Returns					Stance inputs	
		Mean (%)	SD (%)	Skew	Kurt	Vol. ann. (%)	$\bar{P}_t$	$\bar{N}_t$
In-sample (08–18)	2754	−0.014	1.489	−0.063	5.699	23.64	0.306	−0.278
OOS (19–23)	1260	0.033	1.327	−0.968	13.079	21.07	0.318	−0.345
Full sample	4014	0.001	1.440	−0.290	7.424	22.87	0.310	−0.299

3.1 Target variable and volatility measure

The Euro Stoxx 50 index (SX5E) reflects the 50 largest stocks among the Eurozone’s 20 supersectors in terms of free-float market capitalization with a maximum weight of 10%

per constituent. This index was chosen as the target variable since ECB communication is addressed to financial markets across the Eurozone, of which the Euro Stoxx 50 is among the most well known and liquid indices, serving as a benchmark and underlying for financial products and other indices (STOXX, 2024). Daily closing prices are obtained from Yahoo Finance over the sample period. Daily index returns are calculated as

$$r_t = \ln\left(\frac{I_t}{I_{t-1}}\right), \quad (3.1)$$

where I_t is the closing index price on day t and r_t is the daily log return. Squared log returns serve as a proxy for realized daily volatility and are used as the benchmark series against which model forecasts are evaluated.

3.2 News measure

Our news index is constructed from three types of publicly available ECB communication:

1. Introductory statements to the Monetary Policy press conference of the Governing Council (referred to as MPS), published monthly until 2014 and on a six-week cycle thereafter (eight per year). These are the most direct expression of ECB policy intent and the most market-moving ECB communication.
2. Monthly Bulletins, published monthly until 2015, and Economic Bulletins, which replaced the former from 2015 and are published eight times per year (two weeks after each MPS). Bulletins contain ECB staff's detailed economic assessment and risk evaluation in more candid language than the polished MPS.
3. Speeches by the ECB President (Trichet until Oct 2011, Draghi from Nov 2011 to Oct 2019, Lagarde from Nov 2019) and Chief Economist (Stark until Dec 2011, Praet from Jan 2012 to May 2019, Lane from Jun 2019). Speeches are included to capture inter-MPS communications at forums, conferences and other appearances. Other Executive Board speeches are excluded so the resulting series reflects communication that markets treat as authoritative only.

3.2.1 Collection and corpus cleaning

Documents were collected via ECB's FOEDB publication API (ECB, 2024), which underlies their websites and contains metadata as machine-readable JSON. Documents are extracted based on publication date, document type and relevant speaker. A total of 1163 documents is collected, including 826 speeches, 182 MPS, 84 Monthly Bulletins and 71 Economic Bulletins. Q&A transcripts are removed from the MPS documents. Non-content text elements (e.g. postal address, copyright notice, publication imprint, etc.) are stripped. Six slide-deck PDFs falsely catalogued as speeches by the API and one duplicated MPS are dropped. After these filters the corpus contains 1,159 documents, thereof 823 speeches, 181 MPS, 84 Monthly Bulletins, and 71 Economic Bulletins.

3.2.2 Sentence segmentation and cruft filter

Documents are split into sentences using punctuation-based boundary detection, with protection for common abbreviations (Mr., Dr., e.g., i.e., et al., etc.). Each sentence is passed through a deterministic cruft filter and dropped as non-content if any of these categories apply: (i) shorter than 30 characters (e.g., captions, footers), (ii) digit ratio of 0.20 or higher (e.g. table row residue from PDF extraction), and (iii) matching a set of regular expressions that capture non-content items (e.g., chart references, page headers). After Q&A removal (41,100 sentences) and cruft filtering (29,515 sentences), 274,393 sentences remain for scoring.

3.2.3 Scoring

Following Bodilsen and Lunde (2024), Smales (2016) and Sadik et al. (2018), we initially scored sentences based on sentiment using CentralBankRoBERTa, a model trained on central bank communications (Pfeifer and Marohl, 2023). However, CentralBankRoBERTa targets affective sentiment, which is appropriate for general financial news (as used in the reference work) where positive and negative content genuinely varies, but inadequate for the ECB corpus which deliberately blends positive and negative language to manage market expectations. Thus, the initial scores returned near-uniform neutrality across documents, demonstrating that generic sentiment classifiers fail on this corpus. We therefore substituted stance scoring using gt fintechlab/FOMC-RoBERTa (Shah et al., 2023), a transformer fine-tuned on FOMC documents for hawkish/dovish/neutral classification. For each sentence, the model returns a probability vector over the three stance classes. The cross-bank application is motivated by the shared institutional register and vocabulary of these two banks. However, residual differences in vocabulary and policy regime between FOMC and ECB are a measurement risk that we address through the source-decomposition robustness checks (R1.2–R1.4).

3.2.4 Sanity check: identified events

To verify that FOMC-RoBERTa produces directionally valid readings for ECB communication, we inspect document stance values within a ± 7 -day window around six test events. Draghi’s “whatever it takes” speech (ECB, 2012a), the Outright Money Transactions (OMT) announcement (ECB, 2012b), and the COVID Pandemic Emergency Purchasing Programme (PEPP) launch (ECB, 2020) are expected to register as strongly dovish. The first post-COVID rate hike of 25 bp (ECB, 2022a), the 75 bp hike (ECB, 2022b), and the September 2023 terminal-hike statement (ECB, 2023) are expected to register as strongly hawkish. All six case readings carry the expected sign: the three easing events display high dovish and low hawkish scores, the three tightening events the opposite. We treat this check as a necessary condition for proceeding with the stance series.

3.2.5 Document-level aggregation

For each document and each confidence threshold $\tau \in \{0, 0.50, 0.80\}$, sentences are defined as dovish (hawkish) if their argmax class is dovish (hawkish) and the maximum probability

is at least τ . The document-level dovish (hawkish) score is the sum of dovish (hawkish) probabilities over qualifying sentences, divided by the total sentence count of that document. We divide by total document sentence count instead of qualifying (non-neutral) count since the latter mechanically forces the dovish and hawkish share to sum to 1, resulting in perfect structural collinearity of -1 between the two directional series. Dividing by the total count lets the neutral share that does not enter the model absorb variation and unpins the hawkish and dovish shares from each other. We report the main results at the middle threshold $\tau = 0.50$, so we include a sentence's probability mass in the directional score only when the classifier is at least more confident than uniform about its assigned class. The two flanking thresholds, $\tau = 0$ and $\tau = 0.80$ are retained for robustness. At $\tau = 0.50$, correlation is mildly positive at 0.343 indicating that more policy-loaded documents tend to have more of both directional sentence types.

3.2.6 Daily aggregation, persistence and rescaling

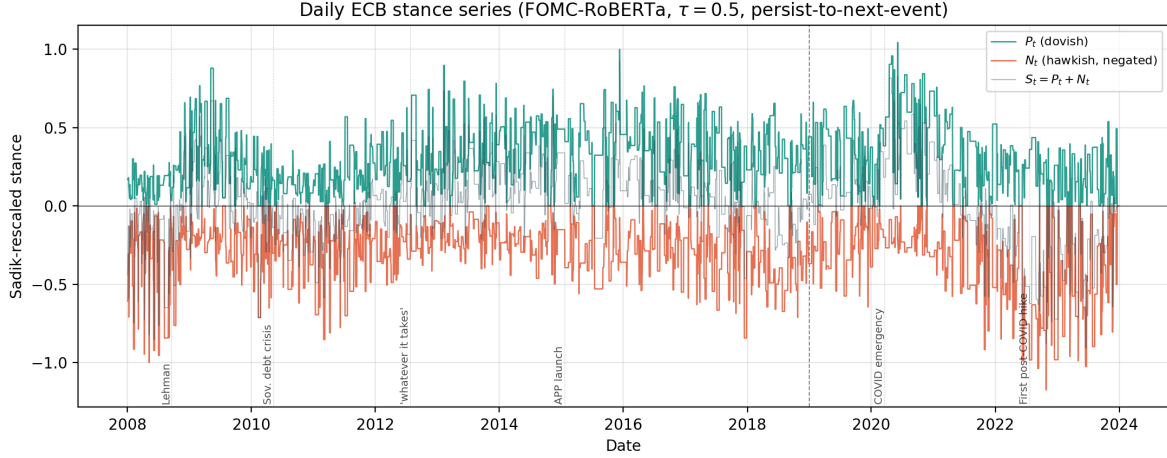
Documents are stamped to a date by publication day. On days with multiple documents, the daily value is an equal-weighted mean across documents to prevent a long Monthly Bulletin from dominating a short speech that markets may, in fact, weigh more heavily. The result is a calendar-day series from Jan 1, 2008 to Dec 31 2023 with a news event on 916 days (MPS / Speeches / Bulletins occur on 157 / 642 / 155 of these days). To produce a continuous daily series given the irregular publication of ECB documents, we let signals persist-to-next-event between event days, rather than the within-day exponential decay of news-impact scores employed by Sadik et al. (2018), which is calibrated for continuously-arriving news and would zero out signal between ECB releases. This preserves the NA-GARCH model's input shape while respecting the lumpy, event-driven nature of ECB communication. Following Sadik et al. (2018), we rescale each persisted directional series by its in-sample (2008-2018) maximum to align ranges across the exponential argument and parameters. The daily model inputs are:

$$P_t = \frac{P^{(d,t,\tau)}}{\max_{s \in \mathcal{I}} P^{(d,s,\tau)}}, \quad N_t = -\frac{H^{(d,t,\tau)}}{\max_{s \in \mathcal{I}} H^{(d,s,\tau)}}, \quad (3.2)$$

where $P^{(d,t,\tau)}$ is the persisted document dovish score on day t , $H^{(d,t,\tau)}$ the corresponding hawkish score and $\mathcal{I}$ denotes the in-sample window. The hawkish term is negated to match NA-GARCH sign convention ($P_t \geq 0, N_t \leq 0$). Dovish is mapped to positive and hawkish to negative since easing typically calms markets and tightening disturbs them. In-sample ranges are $[0, 1]$ for dovish and $[-1, 0]$ for hawkish. Out-of-sample values can exceed those bounds, since the in-sample max is fixed at estimation time. In-sample, the realised maxima are 0.591 for the dovish and 0.541 for the hawkish series. Figure 1 plots the resulting P_t and N_t series; the dovish tilt of 2008–2018 and the hawkish swing in 2022–2023 are visible.

Table 2: In-sample maxima at $\tau = 0.50$.

Source	Dovish max	Hawkish max
All sources	0.5913	0.5414
MPS only	0.5302	0.5414
Speeches only	0.5913	0.3860
Bulletins only	0.4537	0.3290

**Figure 1:** Daily ECB stance at $\tau = 0.50$. Selected reference events: Lehman (Sep08), sovereign debt crisis (2010–12), Draghi speech (Jul12), APP launch (Jan15), COVID emergency (Mar20), first post-COVID rate hike (Jul22).

3.3 Controls

The daily time series in Appendix C, covering the full sample period, are used to compute Bernoth-style residualized stance surprises in robustness checks R1.9 and R1.10. (see Section 4.6). They are not part of the main forecasting model. All series are aligned to the Euro Stoxx 50 trading-day frequency via forward-fill where necessary.

4 Methodology

4.1 Benchmark models

We estimate three benchmark variance specifications, all on the demeaned log-return $\varepsilon_t = r_t - \mathbb{E}[r_t]$ with $\varepsilon_t = \sigma_t z_t$ and z_t i.i.d. Student- t with ν degrees of freedom:

B1.1 – GARCH(1,1) (Bollerslev, 1986):

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (4.1)$$

B1.2 – GJR-GARCH(1,1) (Glosten et al., 1993), allowing the conditional variance to react

asymmetrically to negative shocks via the indicator $I_{t-1} = \mathbf{1}\{\varepsilon_{t-1} < 0\}$:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma_{\text{lev}} I_{t-1} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (4.2)$$

B1.3 – EGARCH(1,1) (Nelson, 1991), modeling $\log \sigma_t^2$ to enforce positivity of the variance without inequality constraints:

$$\log \sigma_t^2 = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma_{\text{lev}} z_{t-1} + \beta \log \sigma_{t-1}^2. \quad (4.3)$$

B1.3 is the primary benchmark in Sadik et al. (2018), and we retain it as ours since it disentangles size and sign effects in the leverage response, which is the closest news-free analogue to what the NA-GARCH news term is designed to capture. We include B1.2 as an additional benchmark because the GJR indicator-based leverage term is the parametrically simplest competitor to the multiplicative news scaling factor of NA-GARCH to test whether the news channel adds anything beyond the asymmetry already captured by sign-conditional shocks. All benchmarks are estimated with the arch Python package (Sheppard (2021)) by Student- t quasi-maximum likelihood.

4.2 News-augmented GARCH

Sadik et al. (2018) augment the standard GARCH(1,1) variance equation by a multiplicative news-driven scaling factor:

$$\sigma_t^2 = f(P_{t-1}, N_{t-1}) (\omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2), \quad f(P, N) = a + \frac{1}{2}b \left[\frac{e^{\kappa P} - 1}{e^{\kappa P} + 1} - \frac{e^{\gamma N} - 1}{e^{\gamma N} + 1} \right] \quad (4.4)$$

The bracketed term is a difference of two $\tanh$ -like functions of the positive and negative news intensities, with direction-specific sensitivities κ and γ . The leading constants a and b control the baseline level and the news-responsiveness of the scaling factor. At the boundary $P = N = 0$, where f collapses to the constant a and the model reduces to a re-scaled GARCH(1,1). We retain Sadik et al.'s functional form unchanged. We replace the news sentiment indices by Sadik et al. (2018) with stance-based analogues: P_t becomes the forward-filled, in-sample-max-rescaled daily dovish probability, and N_t the negated, forward-filled, in-sample-max-rescaled daily hawkish probability (both constructed in Section 3). Under this substitution, parameter interpretations are preserved. We estimate two news-augmented main variants, distinguished by how the directional stance signals enter the news function:

NA1.1 – NA-GARCH-net. The two directional signals are combined into a single net stance input $S_t = P_t + N_t$, on which the news function operates with a single sensitivity parameter κ . This is the most parsimonious news-augmented specification and tests whether the direction of net stance, treated as a scalar, contains information for the conditional variance.

$$\sigma_t^2 = f(S_{t-1}) (\omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2), \quad f(S) = a + \frac{1}{2}b \left[\frac{e^{\kappa S} - 1}{e^{\kappa S} + 1} \right]. \quad (4.5)$$

NA1.2 – NA-GARCH-asym. The directional signals P_t and N_t enter separately, with direction-specific sensitivities κ and γ . This is the original Sadik et al. (2018) specification, allowing the news function to respond more strongly to one stance direction than to the other.

$$\sigma_t^2 = f(P_{t-1}, N_{t-1}) (\omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2), \quad f(P, N) = a + \frac{1}{2}b \left[\frac{e^{\kappa P} - 1}{e^{\kappa P} + 1} - \frac{e^{\gamma N} - 1}{e^{\gamma N} + 1} \right] \quad (4.6)$$

NA1.1 is a more parsimonious specification with one fewer free parameter: it imposes that P and N enter additively through a single net stance before the nonlinearity, rather than each through its own tanh-like response.

Please refer to Appendix E for information regarding an earlier specification.

4.3 In-sample estimation

4.3.1 Likelihood and innovation distribution

All models are estimated by maximum likelihood in-sample. Innovations are assumed Student- t with degrees of freedom ν estimated jointly with the variance parameters. The standardised- t log-likelihood contribution of observation t is

$$\ell_t = \log \Gamma\left(\frac{\nu+1}{2}\right) - \log \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \log [\pi(\nu-2)\sigma_t^2] - \frac{\nu+1}{2} \log \left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right), \quad (4.7)$$

with $\nu > 4$ enforced through the reparameterisation $\nu = 4 + e^{\theta_\nu}$, ensuring that the unconditional fourth moment exists at all interior points. A Gaussian-innovation variant of NA1.2 is retained as a robustness check (R1.7), but is not the primary specification.

4.3.2 Reparameterisation and standard errors

For the NA-GARCH variants, the positivity constraints on $\omega, a, b, \kappa, \gamma$ are imposed through a logarithmic reparameterisation $\theta_x = \log x$ and the optimisation is conducted in unconstrained θ -space, using a multi-restart Nelder–Mead / L-BFGS-B sequence with starting values seeded from a GARCH(1,1) pre-fit on the residuals. Benchmark standard errors come from the arch package’s robust sandwich estimator; for the NA-GARCH variants, we compute a finite-difference numerical Hessian of the negative log-likelihood at the optimum, invert it, and map each parameter’s θ -space standard error through its Jacobian ($\widehat{SE}(x) = x \widehat{SE}(\theta_x)$). Where the Hessian is non-invertible, standard errors are reported as missing.

4.3.3 Residual diagnostics

For every estimated model we report Ljung–Box Q statistics on standardised residuals $\hat{z}_t = \varepsilon_t / \hat{\sigma}_t$ and squared standardised residuals $\hat{z}_t^2$ at lags 5, 10, and 20.

4.4 Out-of-sample forecasting design

4.4.1 Forecast origins and windows

Out-of-sample forecasts are generated at weekly forecast origins, with five trading days between successive origins. This yields 252 forecast origins per (scheme, horizon) over the 2019–2023 OOS window. At every origin the model is re-estimated and a fresh forecast is produced; warm-starting from the previous origin’s MLE is applied for the NA-GARCH variants (where the optimiser benefits substantially), but not for the arch-wrapped benchmarks, which the package refits from scratch. Two windowing schemes are evaluated:

- **Rolling Window (RW):** the training window is fixed at the in-sample length of 2,754 observations, dropping the oldest observation and adding the newest at each origin. This implicitly assumes that the data-generating process drifts and that older information should be discounted.
- **Increasing Window (IW):** the training window starts at the in-sample first observation and grows with each origin to include all data up to that origin. This assumes a stable data-generating process and that more data is always better.

The two schemes bracket the realistic spectrum of analyst behaviour, and the difference between them is itself informative about the stability of variance dynamics over the COVID and post-COVID horizon. The five main models (B1.1, B1.2, B1.3, NA1.1, NA1.2) are evaluated under both schemes for each horizon, giving twenty (model, scheme, horizon) combinations. The robustness variants of Section 4.6 are evaluated under RW only.

4.4.2 Forecast horizons and targets

Two horizons are considered:

$$h = 1 : \quad \hat{\sigma}_{t+1|t}^2 \text{ vs. } \varepsilon_{t+1}^2 \qquad h = 5 : \quad \sum_{k=1}^5 \hat{\sigma}_{t+k|t}^2 \text{ vs. } \sum_{k=1}^5 \varepsilon_{t+k}^2.$$

For NA1.1 and NA1.2, the multi-step variance recursion requires assumptions about future news inputs. We freeze P and N at the most recent observed value over the $h = 5$ horizon, which is the natural counterpart of the persistence rule governing the daily series: in the absence of a new event, the model carries its last reading forward.

4.5 Out-of-sample evaluation

4.5.1 Loss functions

The primary evaluation loss is Patton (2011) robust QLIKE, which is consistent for the conditional-variance ranking under a noisy volatility proxy and is robust to the use of squared

returns as the realised target.

$$\mathcal{L}_t^{\text{QLIKE}} = \log \hat{\sigma}_t^2 + \frac{\varepsilon_t^2}{\hat{\sigma}_t^2}, \quad (4.8)$$

RMSE is reported alongside for completeness but is known to be sensitive to the proxy and is not used as the basis of inference.

4.5.2 Diebold–Mariano tests

For each non-baseline model, we test equal predictive accuracy of its QLIKE loss against the B1.1 GARCH(1,1) baseline using the Diebold–Mariano (1995) statistic

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}(d)/T}}, \quad d_t = \mathcal{L}_t^{\text{cand}} - \mathcal{L}_t^{\text{B1.1}}. \quad (4.9)$$

For $h = 1$, $\hat{V}(d)$ is the sample variance of the loss differential. For $h = 5$, we use the Newey–West (1987) HAC estimator with a Bartlett kernel and bandwidth $h - 1 = 4$, which is the standard truncation for h -step forecasts whose loss differentials inherit at most $h - 1$ lags of correlation. Two-sided p -values follow from asymptotic normality of the studentised statistic. Tests are run within each (scheme, horizon). We additionally report DM comparisons against the B1.2 GJR baseline, since this is the model that empirically dominates in the Section 4.5.3 confidence set.

4.5.3 Model confidence set

To control the joint multiple-testing risk across the main models, we compute the Model Confidence Set of Hansen et al. (2011) within each (scheme, horizon). The MCS uses the equivalence statistic $T_{\max}$, computed by stationary bootstrap on the QLIKE losses with 5,000 replications, and iteratively eliminates models that are statistically dominated. The reported $\widehat{\text{MCS}}_{90\%}$ retains the models that cannot be eliminated at confidence level $1 - \alpha = 0.90$.

4.5.4 Forecast combination

As a sharper test of orthogonal forecasting content, we estimate the optimal linear combination of NA1.2 with B1.2 (GJR-GARCH), i.e., the dominant benchmark in the Model Confidence Set of Section 4.5.3, by grid-searching the weight $w \in [0, 1]$ in increments of 0.05 (21 grid points) for $(1 - w) \cdot \text{B1.2} + w \cdot \text{NA1.2}$. A value of w^* close to zero indicates that NA1.2 carries no information orthogonal to GJR; a value materially above zero would indicate that, even if NA1.2 is dominated stand-alone, its forecast errors are diversifying relative to GJR's. This is the most demanding standard the news term can be held to.

4.5.5 Sub-sample evaluation

The OOS window contains three economically distinct regimes that motivate a regime-conditional evaluation: a 2019 *normal* period (50 observations before the COVID shock), a

2020–2021 COVID period (101 observations covering the pandemic shock and recovery), and a 2022–2023 inflation period (101 observations covering the tightening cycle). QLIKE and RMSE are recomputed within each sub-sample, allowing us to check whether any model’s apparent overall performance is driven by a single regime, and whether the news-augmented specifications outperform during regimes in which one might *a priori* expect them to: monetary-policy-news-intensive episodes.

4.5.6 Economic significance of the news term

To translate the estimated news block into an economically interpretable quantity, we compute the implied news scaling factor $f(P, N)$ at the four corners of its support box $(P, N) \in [0, 1] \times [-1, 0]$ under the fitted NA1.2 parameters: $(0, 0)$ (no news baseline), $(1, 0)$ (maximum dovish), $(0, -1)$ (maximum hawkish), and $(1, -1)$ (maximum mixed). The range $f_{\max} - f_{\min}$, expressed as a percentage of the no-news baseline $f(0, 0)$, summarises the magnitude of variance amplification the news term is permitted to produce under the in-sample estimates.

4.6 Robustness suite

Ten additional specifications are estimated for robustness. Together they probe the sensitivity of the result to proxy choice, communication channel, confidence threshold, innovation distribution, and three distinct constructions of a stance-based surprise.

R1.1: Alternative volatility proxy. The NA1.2 OOS forecasts are re-evaluated against $|\varepsilon_t|$ in place of ε_t^2 , with the variance forecasts converted to volatility scale via $\sqrt{\hat{\sigma}_t^2}$ for consistency. This tests sensitivity of the QLIKE ranking to the realised-variance proxy.

R1.2–R1.4: Source decomposition. NA1.2 is re-estimated and re-forecast three times with the news inputs restricted to a single communication channel: MPS only (R1.2), speeches only (R1.3), or bulletins only (R1.4). The three variants jointly decompose the all-source signal into formal-decision (MPS), inter-meeting (speeches), and staff-assessment (bulletins) channels.

R1.5: NA-GARCH-asym, $\tau = 0$. NA1.2 is re-estimated in-sample with the stance series computed without any confidence threshold on sentence-level classifier outputs.

R1.6: NA-GARCH-asym, $\tau = 0.80$. NA1.2 is re-estimated with the stance series computed at high confidence only. R1.5 and R1.6 together verify that the main result is not an artefact of the chosen confidence cut.

R1.7: NA-GARCH-asym, Gaussian innovations. NA1.2 is re-estimated with $z_t \sim \mathcal{N}(0, 1)$, to confirm that the Student- t specification is appropriate in the main model.

R1.8: MPS-vs-baseline surprises. The levels-based inputs of the main suite follow Sadik et al. (2018), who treat news as exogenous random impulses priced *ex post*. ECB events, however, are scheduled and partly anticipated, so stock prices reflect expectations of future communication *ex ante*. To test whether the null is an artefact of using stance levels rather than surprises, we construct an event-driven surprise as the deviation of the announced MPS stance from a decay-weighted mean of intra-period speeches and bulletins, decayed forward at the same rate (5-trading-day half-life) between MPS events and absolute-valued to preserve the NA-GARCH sign convention. The full construction is given in Appendix D. The resulting surprise inputs replace the level inputs in NA1.2. All other modelling choices are held fixed.

R1.9: Bernoth surprises with AR(2) lags. Following Bernoth (2025), we treat the unanticipated component of stance as the residual from an OLS regression of P_t and N_t on contemporaneous financial-state controls and two own lags, estimated once on the in-sample event days. The fitted residuals are truncated to the NA-GARCH supports ($P_t \geq 0, N_t \leq 0$), forward-filled under the persist-to-next-event convention, and rescaled by their in-sample maxima exactly as the level series. Appendix D gives the full specification. The rescaled surprise series then replace the level inputs in NA1.2.

R1.10: Bernoth surprises without lags. Symmetric to R1.9 with the AR(2) lag terms dropped, to isolate the surprise component relative to the financial state without conditioning on the bank's own recent tone.

Among the robustness variants, R1.1 reuses NA1.2's OOS forecasts, R1.2–R1.4 are estimated in-sample and forecast under the RW scheme at $h = 1$ and $h = 5$. The threshold, innovation-distribution, and surprise-encoding variants (R1.5–R1.10) are evaluated in-sample only, as their purpose is to test the stability of the in-sample MLE ranking rather than the OOS forecast.

5 Results

Within our empirical design, the ECB stance does not deliver a statistically detectable improvement in out-of-sample variance forecasts over the GARCH benchmark, and is significantly outperformed by GJR in every scheme and horizon. The leverage effect captured parametrically by GJR absorbs the asymmetric variance response that the ECB stance series was designed to add information to.

5.1 In-sample fit

Table 3 reports maximum-likelihood estimates of the five main specifications on the in-sample window (2,754 trading days, 2008–2018). The asymmetric benchmarks dominate the news-augmented variants on both log-likelihood and AIC: EGARCH (B1.3) attains the highest log-likelihood at 8,270.7 (AIC: $-16,531.4$), followed by GJR (B1.2) at 8,253.4 (AIC: $-16,496.9$).

Both news-augmented variants underperform: NA1.1 reaches only 8,193.2 (AIC: $-16,372.3$), essentially identical to the plain GARCH baseline (B1.1) at 8,193.0 (AIC: $-16,378.1$), and NA1.2 reaches 8,195.0 (AIC: $-16,374.0$), a 2-point log-likelihood gain over plain GARCH for four additional parameters, which AIC penalises into a worse score. The leverage effect captured by B1.2 and B1.3 contributes roughly $+60$ to $+77$ log-likelihood units relative to the symmetric GARCH baseline, while the news term in either NA-GARCH variant contributes at most $+2$. Appendix B plots the in-sample fitted conditional volatilities.

Table 3: In-sample MLE estimates, main suite. Lower AIC is better.

Model	N	Log-lik.	AIC	Params
B1.1	2754	8193.03	$-16\,378.06$	4
B1.2	2754	8253.45	$-16\,496.89$	5
B1.3	2754	8270.71	$-16\,531.43$	5
NA1.1	2754	8193.16	$-16\,372.33$	7
NA1.2	2754	8194.98	$-16\,373.96$	8

5.2 Parameter estimates

Parameter estimates for NA1.2 (Table 4) clarify why the news term fails on AIC. The baseline scaling constant $\hat{\alpha} = 0.983$ (s.e. 0.006) is essentially equal to one, i.e., the no-news scaling factor $f(0, 0)$ leaves the GARCH variance recursion approximately unchanged. The dovish sensitivity $\hat{\kappa} = 0.086$ has a standard error of 0.247 and a t -statistic of 0.35, not distinguishable from zero. The hawkish sensitivity $\hat{\gamma} = 3.012$ converges to an interior point in the log-parameter space in which the optimiser operates ($\log \hat{\gamma} \approx 1.10$, well within the bound $\log \gamma \in [-5, 2]$), but the numerical Hessian along the γ direction is near-singular, indicating that the in-sample likelihood is locally flat in γ . Hence, the data does not identify the hawkish sensitivity. Hessian inversion either fails or returns a degenerate variance for γ under the delta-method mapping back to natural-parameter space, so its standard error is reported as missing in Table 4. Only one of the two directional sensitivities is interior-identified, and it is statistically indistinguishable from zero. By contrast, the leverage coefficients in B1.2 ($\hat{\gamma}_{lev} = 0.191$, $t = 5.65$) and B1.3 ($\hat{\gamma}_{lev} = -0.163$, $t = -7.93$) are large in magnitude and tightly estimated. Note further that in GJR the symmetric ARCH coefficient collapses to $\hat{\alpha} = 0.000$ while the leverage term carries the full shock response ($\hat{\gamma}_{lev} = 0.191$, $t = 5.65$). Hence, the entire conditional-variance response to lagged shocks is sign-conditional. This is precisely the channel through which a news term would have to provide incremental information, and its empirical dominance in the benchmark fit anticipates the headline OOS result.

5.3 Residual diagnostics

Ljung–Box Q -statistics on standardised residuals and squared standardised residuals (Appendix F) yield p -values above 0.10 at every lag (5, 10, 20) for every main-suite model. No specification leaves systematic autocorrelation in either the conditional mean or the condi-

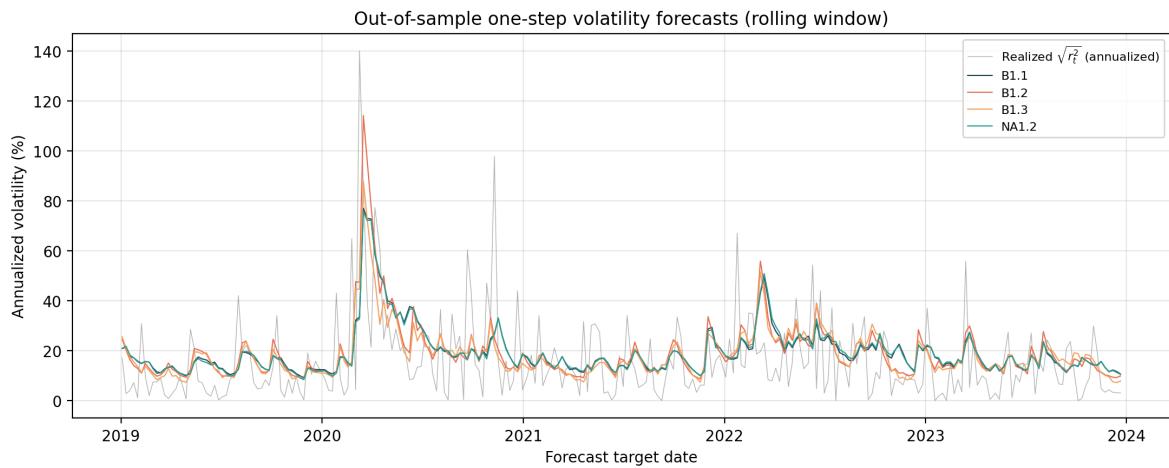
Table 4: In-sample parameter estimates (main suite). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Parameter	B1.1	B1.2	B1.3	NA1.1	NA1.2
ω	0.000** (0.000)	0.000*** (0.000)	-0.175*** (0.003)	0.000*** (0.000)	0.000** (0.000)
α	0.083*** (0.014)	0.000 (0.012)	0.118*** (0.017)	0.084	0.078
β	0.912*** (0.014)	0.897*** (0.020)	0.980*** (0.005)	0.911	0.918
γ_{lev}		0.191*** (0.034)	-0.163*** (0.021)		
a				1.001*** (0.006)	0.983*** (0.006)
b				0.007	0.092
κ				0.007	0.086 (0.247)
γ					3.012
ν	6.769*** (0.877)	7.512*** (1.081)	7.867*** (1.191)	6.747*** (0.853)	6.712*** (0.844)
N	2754	2754	2754	2754	2754
Log-lik.	8193.0	8253.4	8270.7	8193.2	8195.0
AIC	-16378.1	-16496.9	-16531.4	-16372.3	-16374.0

tional variance, so the AIC ranking is not contaminated by mis-specified short-run dynamics. The relative under-performance of the NA-GARCH variants reflects genuine inadequacy of the news term as a signal, not an artefact of model misfit.

5.4 Out-of-sample forecast accuracy

Figure 2 plots one-step forecasts under the rolling-window scheme against realised volatility. The four model forecasts (B1.1, B1.2, B1.3, NA1.2) are visually indistinguishable except around the March 2020 and Q1 2022 volatility spikes, where the asymmetric benchmarks (B1.2, B1.3) move more aggressively than the news-augmented NA1.2.

**Figure 2:** OOS one-step annualised volatility forecasts under RW scheme, against realised $\sqrt{r_t^2}$.

Appendix G reports the full QLIKE and RMSE table. The in-sample ranking is preserved out of sample: EGARCH attains the lowest QLIKE at every horizon and under both schemes (-7.767 at RW, $h = 1$), followed by GJR (-7.758). The news-augmented specifications are bunched around the plain-GARCH baseline (-7.655 at RW, $h = 1$): NA1.1 is numerically indistinguishable from B1.1, while NA1.2 is fractionally worse than B1.1 across all four (scheme, horizon) combinations. Switching to RMSE does not overturn the substantive conclusion: the news-augmented specifications do not dominate the asymmetric benchmarks. The gap between asymmetric benchmarks and news-augmented variants is large enough to be visible in the raw numbers.

5.4.1 Diebold-Mariano tests

Diebold–Mariano tests against the B1.1 GARCH baseline (Appendix H) reveal that B1.2 (GJR) beats B1.1 at the 5% level in every (scheme, horizon) cell (p between 0.025 and 0.043), and B1.3 (EGARCH) does so at one out of four cells with the remaining three in the 5–10% range. Neither news-augmented variant is statistically distinguishable from B1.1: NA1.1 and NA1.2 yield p -values uniformly above 0.10 in all eight tests, and the loss differential for NA1.2 is in fact positive in every cell, meaning NA1.2 produces higher QLIKE than the plain-GARCH baseline at all horizons. Both NA1.1 and NA1.2 are statistically worse than the asymmetric B1.2 at $p < 0.05$ in all four cells, as are the source-decomposition robustness variants R1.2–R1.4. The news term does not merely fail to improve forecasts; it is statistically dominated by the asymmetric-shock benchmark across every evaluation cell.

5.4.2 Model Confidence Set

The Hansen et al. (2011) Model Confidence Set at the 90% confidence level (Table 5) retains only B1.2 and B1.3 in every (scheme, horizon) combination. The plain-GARCH baseline and both news-augmented variants are eliminated as statistically inferior under both schemes and at both horizons. Under the rolling-window scheme the source-decomposition robustness variants (R1.2 MPS-only, R1.3 speeches-only, R1.4 bulletins-only) are also excluded, confirming that no communication-channel decomposition rescues the news term. The MCS thus provides a multiple-testing-corrected statement of the headline result: the conditional-variance information set that beats GARCH is captured entirely by sign-conditional shocks, with no incremental contribution from stance-derived ECB news.

Table 5: Hansen et al. (2011) Model Confidence Set at 90% confidence.

Scheme	h	T	In MCS	Excluded
RW	1	252	B1.2, B1.3	B1.1, NA1.1, NA1.2, R1.2, R1.3, R1.4
RW	5	252	B1.2, B1.3	B1.1, NA1.1, NA1.2, R1.2, R1.3, R1.4
IW	1	252	B1.2, B1.3	B1.1, NA1.1, NA1.2
IW	5	252	B1.2, B1.3	B1.1, NA1.1, NA1.2

5.4.3 Forecast combination

A stronger test than the DM statistic is whether NA1.2's forecast errors are at least diversifying with respect to B1.2, i.e., whether NA1.2 contributes orthogonal information even if it is dominated on a stand-alone basis. Table 6 reports the optimal weight w^* minimising QLIKE for the linear combination $(1 - w) \cdot \text{B1.2} + w \cdot \text{NA1.2}$. In all four (scheme, horizon) combinations $w^* = 0$ exactly. The exact-zero result means QLIKE is strictly increasing in w from $w = 0$ onward across all four cells: the loss function has no interior minimum, and the optimum sits at the corner where the NA-GARCH contribution is zero.

Table 6: Forecast combination test

Scheme	h	Model A	Model B	T	QLIKE_A	w^*	QLIKE_{comb}	ΔQLIKE
RW	1	B1.2	NA1.2	252	-7.7576	0.0000	-7.7576	0.0000
RW	5	B1.2	NA1.2	252	-6.3667	0.0000	-6.3667	0.0000
IW	1	B1.2	NA1.2	252	-7.7581	0.0000	-7.7581	0.0000
IW	5	B1.2	NA1.2	252	-6.3653	0.0000	-6.3653	0.0000

5.5 Economic significance of the news term

To translate the failure of the news term into an economic magnitude, Table 7 reports the implied news scaling factor $f(P, N)$ at the four corners of its support box under the fitted NA1.2 parameters. The baseline $f(0, 0) = 0.983$, the maximum dovish $f(1, 0) = 0.985$, the maximum hawkish $f(0, -1) = 1.025$, and the maximum mixed $f(1, -1) = 1.027$. The range $f_{\max} - f_{\min} = 0.044$ means a 4.46% deviation from the no-news baseline. Even at the most extreme news intensities the in-sample window ever realised, the variance scaling moves by less than 5% relative to the no-news case. The estimated news block is not merely statistically weak, the magnitude it is permitted to produce, at the in-sample MLE, is too small to matter for any practical forecasting application. Figure 3 plots the fitted $f(P, N)$ surface across the in-sample support box. Variation is dominated by the hawkish axis N_t ; the dovish axis P_t contributes near-zero slope. Total range across the support box: 4.46% of the no-news baseline.

Table 7: Economic significance of the news effect under fitted NA1.2

Quantity	Value
$f(0, 0)$ baseline	0.9829
$f(1, 0)$ max. dovish	0.9849
$f(0, -1)$ max. hawkish	1.0247
$f(1, -1)$ max. mixed	1.0267
$f_{\max}$	1.0267
$f_{\min}$	0.9829
$f_{\max} - f_{\min}$ (% of baseline)	4.46%

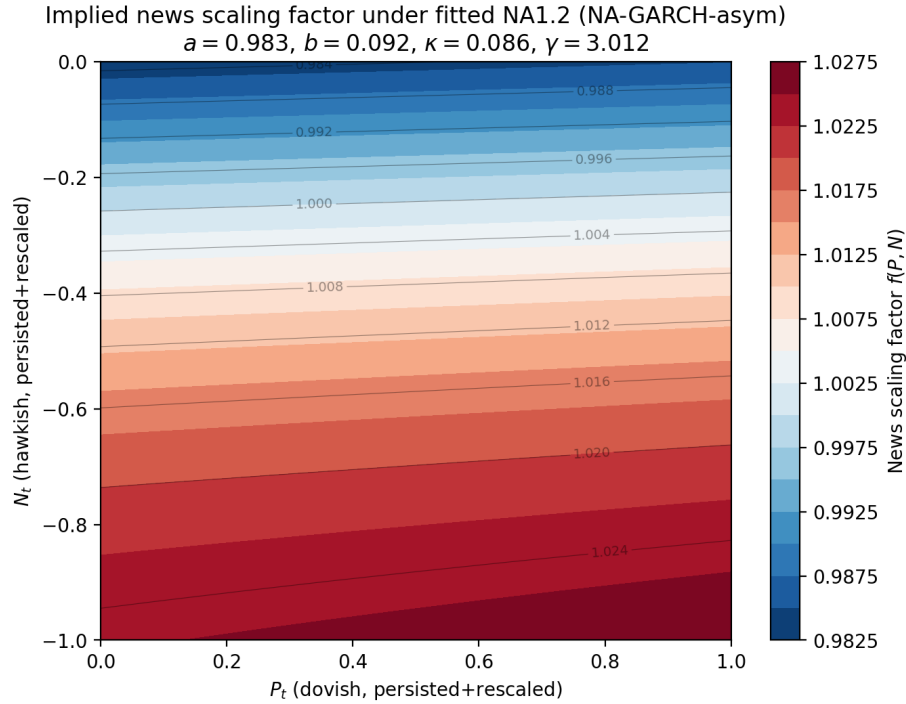


Figure 3: Implied news scaling factor $f(P_t, N_t)$ under fitted NA1.2 parameters ($\hat{a} = 0.983$, $\hat{b} = 0.092$, $\hat{\kappa} = 0.086$, $\hat{\gamma} = 3.012$), plotted across the in-sample support $(P, N) \in [0, 1] \times [-1, 0]$.

5.6 Sub-sample evaluation

If the news effect were time-varying, e.g. present during monetary-policy-intensive episodes but masked in calm periods, the overall null might mask a conditional effect. Appendix I reports QLIKE by sub-sample: 2019 normal (50 origins), 2020–2021 COVID (101 origins), and 2022–2023 inflation (101 origins). The ranking is identical across all three regimes: EGARCH and GJR form the statistically superior benchmark set, NA-GARCH variants cluster around the GARCH baseline. There is no regime in which the news term outperforms the asymmetric benchmarks, including the 2022–2023 tightening cycle, in which one might expect ECB stance to carry maximal forecasting power. The Smales (2016)-style time-varying-news hypothesis is not supported by this corpus.

5.7 Robustness suite

The source-decomposition variants R1.2–R1.4 (Table 9) show that no single communication channel (only MPS, speeches or bulletins) produces an in-sample log-likelihood improvement over the all-source NA1.2. In fact MPS-only and speeches-only fall below the plain-GARCH baseline. The confidence-threshold variants R1.5 ($\tau = 0$) and R1.6 ($\tau = 0.80$) yield log-likelihoods within 0.35 of NA1.2, confirming the result is not driven by the choice of threshold. The Gaussian-innovations variant R1.7 underperforms its Student- t counterpart by 54 log-likelihood units (AIC gap of 108), validating the Student- t choice for the main specification. The alternative volatility-proxy check (Table 8) re-evaluates NA1.2's OOS forecasts against $|\varepsilon_t|$

on the volatility scale rather than ε_t^2 on the variance scale. The QLIKE ranking is preserved under this proxy substitution, so the null is not an artefact of the squared-return target. None of the three surprise-encoding variants rescue forecasting power: the MPS-vs-intra-period-baseline construction (R1.8) yields $\Delta\ell = -0.07$ relative to NA1.2, which is economically and statistically zero. The Bernoth-style residualisation with AR(2) lags (R1.9) yields $\Delta\ell = -1.45$, and the no-lags variant (R1.10) yields $\Delta\ell = -1.64$. All three alternative input constructions underperform raw levels, and the null result is robust across all three definitions of surprise.

Table 8: Robustness to volatility-proxy choice

Scheme	h	T	QLIKE	RMSE
<i>R1.1</i> $ r_t $ target (volatility scale)				
RW	1	252	-3.6695	0.0104
RW	5	252	-2.0541	0.0303
IW	1	252	-3.6643	0.0105
IW	5	252	-2.0654	0.0300

Table 9: In-sample MLE estimates, robustness suite. R1.2–R1.10

Model	Description	Dist.	T	ℓ	AIC	$\Delta\ell$	ΔAIC
R1.2	MPS-only stance source	t	2749	8179.60	-16343.19	-15.38	+30.77
R1.3	Speeches-only stance source	t	2752	8188.97	-16361.94	-6.01	+12.03
R1.4	Bulletins-only stance source	t	2754	8193.17	-16370.33	-1.82	+3.63
R1.5	No confidence threshold ($\tau = 0$)	t	2754	8195.21	-16374.43	+0.23	-0.47
R1.6	High-confidence only ($\tau = 0.80$)	t	2754	8195.29	-16374.57	+0.31	-0.61
R1.7	Gaussian innovations	$\mathcal{N}$	2754	8140.24	-16266.48	-54.74	+107.48
R1.8	MPS surprise vs. baseline	t	2754	8195.22	-16374.43	-0.07	-0.47
R1.9	Bernoth surprises, AR(2) lags	t	2752	8188.23	-16360.45	-1.45	+2.90
R1.10	Bernoth surprises, no AR lags	t	2754	8193.65	-16371.31	-1.64	+3.27

6 Limitations and future work

We identify three limitations of the present study that warrant discussion. First, the stance classifier (FOMC-RoBERTa) is FOMC-trained. Sanity checks in Section 3.2.4 confirm directional validity on ECB events, but level magnitudes may be miscalibrated for ECB-specific instruments. Bernoth (2025) develops an ECB-fine-tuned classifier, which was however not publicly available for download at the time of writing and producing an equivalent fine-tuned classifier was outside the scope of this paper. Replicating the study with it is left to future work. Second, none of the three surprise-encoding constructions in the robustness suite (Section 4.6) achieves a clean separation between a pure *communication* surprise and a *policy-decision* surprise on MPS days; all three operate on the post-decision stance level. Identification strategies that condition on high-frequency rate surprises (Gürkaynak et al.,

2005) or use intraday event-study methodologies (Altavilla et al., 2019) would allow a sharper test of whether the communication channel, net of the policy action, carries incremental volatility information. This is left for future work. Third, this study treats ECB communication as the sole source of macroeconomic news. In practice, market expectations are also shaped by broader macro-financial developments reported outside the ECB corpus like geopolitical events or energy shocks which affect both realised volatility and the ECB's own communication stance. Because these external shocks enter the model only through their lagged effect on ε_t^2 and σ_t^2 , the stance series and the volatility target are not fully separable sources of information. Incorporating a broader macro-news index alongside the ECB stance series, or conditioning on realised measures of global uncertainty, would provide a more complete test of the incremental value of institutional communication.

7 Conclusion

ECB communication does not improve out-of-sample volatility forecasts for the Euro Stoxx 50 beyond standard asymmetric GARCH benchmarks. GJR-GARCH and EGARCH consistently outperform the News-Augmented GARCH extensions across all model specifications, horizons, and estimation schemes. The null is robust to variation in classifier confidence thresholds, communication source decompositions, volatility proxies, and three distinct surprise-encoding constructions. A structural interpretation consistent with the null is that ECB communication is scheduled, anticipated, and continuously monitored by professional analysts, so much of its informational content is incorporated into asset prices before official publication. Testing this directly via high-frequency rate-surprise residuals (Gürkaynak et al., 2005; Altavilla et al., 2019) is left for future work. The asymmetric volatility dynamics already captured by GJR, in particular the leverage effect of negative return shocks, appear to absorb whatever variance-relevant signal ECB stance might carry, leaving no incremental contribution for the news term. The paper makes two contributions. First, it extends the NA-GARCH framework to ECB communication using transformer-based stance classification, providing a systematic out-of-sample evaluation of this approach for euro-area equity volatility. Second, the null is informative in its own right: the forecasting gains documented by Bodilsen and Lunde (2024) for commercial macro-news feeds do not generalise to institutional central-bank communication, pointing to anticipation and the structural differences between scheduled policy releases and unscheduled news as the likely mechanism. Communication clearly moves markets; this does not imply that it improves variance forecasts out-of-sample.

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A Euro Stoxx 50 returns and realised volatility

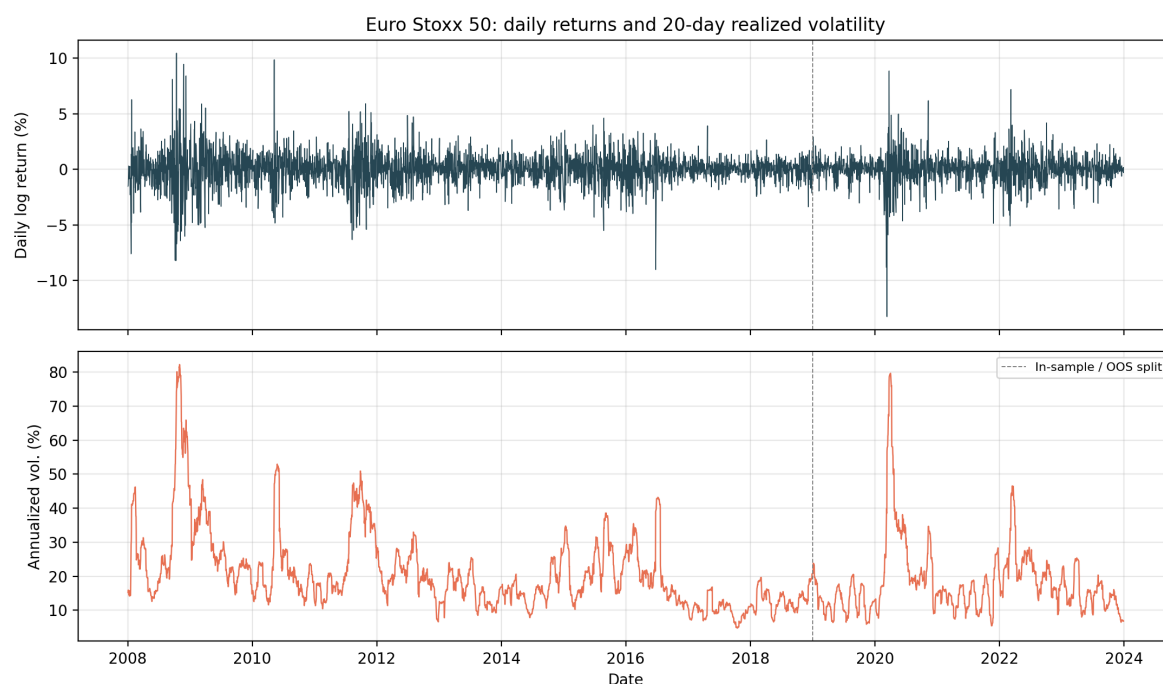


Figure 4: Euro Stoxx 50 daily log returns (top) and 20-day annualised realised volatility (bottom), 2007–2023. The dashed vertical line marks the in-sample/OOS split. Returns exhibit pronounced volatility clustering throughout, with the 2008–09 financial crisis, 2011–12 sovereign-debt crisis, and March 2020 COVID shock the three largest realised-volatility events.

B In-sample fitted volatility

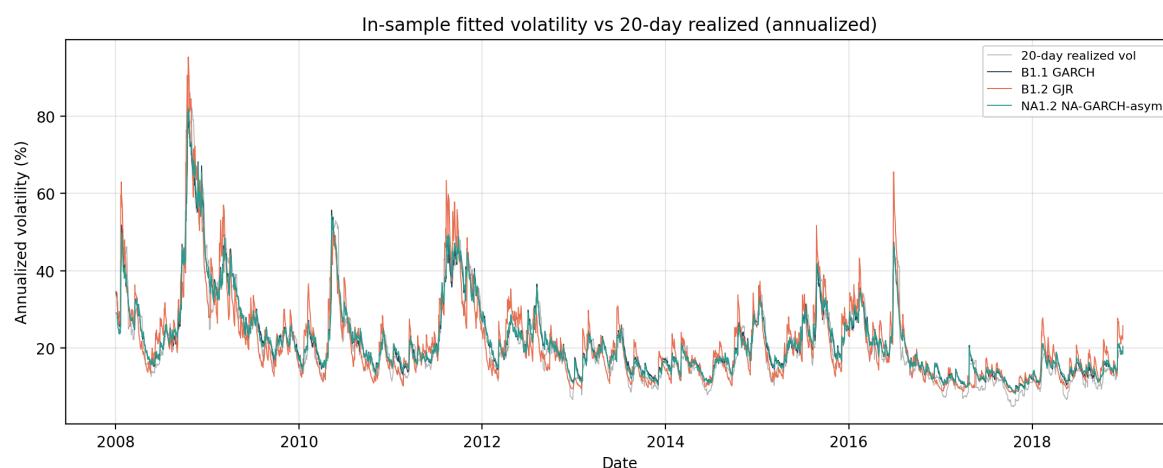


Figure 5: In-sample fitted annualised conditional volatility, 2008–2018: B1.1 GARCH, B1.2 GJR, and NA1.2 NA-GARCH-asym against 20-day realised volatility (grey). Fitted series are visually similar; the asymmetric benchmark (B1.2) tracks high-volatility episodes more tightly than the symmetric GARCH and NA-GARCH variants.

C Financial-state control variables

Variable	Description	Source
VSTOXX	Implied volatility of Euro Stoxx 50 options reflecting forward-looking risk perception in European equity markets	Yahoo Finance
EUR/USD	Euro-dollar exchange rate as proxy for FX risk premia and global risk sentiment	Yahoo Finance
VIX	Implied volatility of S&P 500 options as global risk-aversion proxy with documented spillovers into European volatility	Yahoo Finance
Brent	Front-month Brent crude price. Energy is a major inflation driver, so Brent variation correlates with the inflation outlook the ECB responds to	Yahoo Finance
ECB AAA 10y	Long end of ECB AAA-rated euro-area sovereign yield curve as proxy for term premium and long-run inflation/growth expectations	ECB Data Portal
ECB AAA 1y	Short end of the same curve as proxy for near-term policy rate expectations conditional on the current market state	ECB Data Portal

D Surprise-encoding constructions

R1.8: MPS-vs-intra-period baseline surprises

For each MPS at date t_m with previous MPS at t_{m-1} , let $\mathcal{I}_m$ be the set of non-MPS documents (speeches and bulletins) published in the open interval (t_{m-1}, t_m) . The expected stance heading into the MPS is the decay-weighted mean of intra-period document stance, and the per-event surprise is the deviation of the announced stance from that expectation:

$$\Delta P_{t_m} = P^{(t_m, \tau)} - \sum_{i \in \mathcal{I}_m} w_i P^{(t_i, \tau)}, \quad (\text{D.1})$$

$$\Delta N_{t_m} = (-H^{(t_m, \tau)}) - \sum_{i \in \mathcal{I}_m} w_i (-H^{(t_i, \tau)}), \quad (\text{D.2})$$

with weights

$$w_i = \frac{\exp\{-\lambda(t_m - t_i)\}}{\sum_{j \in \mathcal{I}_m} \exp\{-\lambda(t_m - t_j)\}}, \quad \lambda = \frac{\ln 2}{5}, \quad (\text{D.3})$$

corresponding to a 5-trading-day half-life. The per-event surprise decays forward at the same rate between MPS events and is absolute-valued to match the NA-GARCH sign convention:

$$P_t^{\text{surp}} = |\Delta P_t e^{-\lambda \delta_t}|, \quad N_t^{\text{surp}} = -|\Delta N_t e^{-\lambda \delta_t}|, \quad (\text{D.4})$$

where δ_t is the trading-day count since the most recent MPS event.

R1.9: Bernoth-style residualisation with AR(2) lags

For each event day t within the in-sample window 2008–2018 we estimate the first-stage OLS regressions

$$P_t = \alpha^P + \beta^{P\top} \mathbf{x}_t + \phi_1^P P_{t-1} + \phi_2^P P_{t-2} + \varepsilon_t^P, \quad (\text{D.5})$$

$$N_t = \alpha^N + \beta^{N\top} \mathbf{x}_t + \phi_1^N N_{t-1} + \phi_2^N N_{t-2} + \varepsilon_t^N, \quad (\text{D.6})$$

where $\mathbf{x}_t$ is the vector of contemporaneous daily financial-state controls. Stance lags enter on the persisted series so that they are available as of trading day $t - 1$, a forecast-compatibility requirement. Coefficients are estimated once on the in-sample event days and held fixed thereafter; the same coefficients are then used to compute residuals on every event day in the full sample, mirroring the in-sample-only rescaling logic applied to the level series in the main specification. The raw residuals are truncated to the NA-GARCH supports,

$$P_t^{\text{bernoth}} = \max(0, \hat{\varepsilon}_t^P), \quad N_t^{\text{bernoth}} = \min(0, \hat{\varepsilon}_t^N), \quad (\text{D.7})$$

forward-filled across non-event trading days under the persist-to-next-event convention, and rescaled by their in-sample maxima so the resulting series occupy $[0, 1]$ and $[-1, 0]$ in-sample.

E H_t regime-indicator specification (rejected)

Motivation

Smales (2016) documents that the relationship between news sentiment and equity-implied volatility is time-varying and amplified during periods of market stress. In calmer regimes the marginal news item moves implied volatility less than in stressed regimes, and the asymmetry between positive and negative news widens during crises. A natural alternative to the constant-effect NA-GARCH-asym specification of Section 4 is therefore one in which the news-scaling factor switches on, or amplifies, only when the conditional volatility itself is elevated. Under this alternative, our headline null could mask a regime-conditional effect: ECB stance may matter, but only when markets are already stressed.

Specification

We augment NA1.2 with a binary regime indicator H_t multiplying the news block:

$$\sigma_t^2 = \left[a + \frac{1}{2}(b + \delta H_{t-1}) \cdot (\tanh(\kappa P_{t-1}/2) - \tanh(\gamma N_{t-1}/2)) \right] \cdot (\omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2), \quad (\text{E.1})$$

where

$$H_{t-1} = \mathbf{1}\{\hat{\sigma}_{t-1}^{2,\text{roll}} > c \cdot \bar{\sigma}^2\}, \quad (\text{E.2})$$

$\hat{\sigma}_{t-1}^{2,\text{roll}}$ is the 20-day rolling realised variance, and $\bar{\sigma}^2$ its in-sample long-run mean. The parameter δ captures the regime-conditional amplification of the news scaling: $\delta > 0$ means stance has a larger volatility impact in stressed regimes, $\delta = 0$ collapses the specification back to NA1.2. The threshold c defines what counts as “stressed.” The null of no regime-conditional effect is $H_0 : \delta = 0$.

Inference

Standard χ^2 inference on the likelihood-ratio statistic for $H_0 : \delta = 0$ is invalid because the threshold c is unidentified under the null — when $\delta = 0$, the value of c does not appear in the likelihood and cannot be recovered. This is the Davies (1987) nuisance-parameter problem and inflates the distribution of the LR statistic above its nominal χ_1^2 . We therefore use the Hansen (1999) fixed-regressor bootstrap, which generates the correct empirical distribution of the LR statistic under the null by resampling residuals while holding the regressors and threshold-determining series fixed.

Implementation and results

We grid-search $c \in \{1.0, 1.1, \dots, 2.0\}$ (so the regime ranges from “any above-mean vol period” at $c = 1.0$ to “vol at least $2\times$ long-run mean” at $c = 2.0$), fit equation (E.1) by Student- t MLE at each c , and retain the maximum likelihood across the grid. The maximised log-likelihood exceeds that of NA1.2 by at most 1.06 units (observed $LR = 2.13$), well below the χ_1^2 95th-percentile critical value of 3.84 that one would use unthinkingly.

The fixed-regressor bootstrap with $B = 1,000$ replications returns a 95th-percentile critical value of 12.82 — substantially above the naive χ^2 threshold, reflecting the Davies-problem inflation — and a bootstrap p -value of 0.46. The regime-conditional effect is not statistically distinguishable from zero by either the naive or the correctly-sized test. We additionally tested a continuous “hockey-stick” variant

$$H_{t-1}^{\text{hockey}} = \max\{0, \hat{\sigma}_{t-1}^{2,\text{roll}} / \bar{\sigma}^2 - c\},$$

which avoids the binary-jump discontinuity at the threshold and allows H to grow continuously with how far rolling vol exceeds $c \cdot \bar{\sigma}^2$. The conclusion is unchanged: the maximised log-likelihood gain over NA1.2 is essentially zero and the bootstrap rejects $\delta \neq 0$.

Interpretation

The Smales-style time-varying-news hypothesis is not supported on this corpus. Combined with the regime-disaggregated OOS evaluation in Section 5.6, where NA1.2 fails to outperform the asymmetric benchmarks even within the 2022–2023 monetary-tightening window (the period of maximal ECB policy salience), the evidence points to a flat news-stance effect that is not rescued by conditioning on the volatility regime. Accordingly, the regime indicator is dropped from the main specification.

F Ljung-Box test results

Table 10: Ljung–Box test on (squared) standardised residuals (main suite). H_0 : no autocorrelation.

Model	Lag	Standardised residuals		Squared standardised residuals	
		Q	p -value	Q	p -value
B1.1	5	8.541	0.129	3.741	0.587
	10	11.813	0.298	6.483	0.773
	20	18.053	0.584	17.126	0.645
B1.2	5	8.232	0.144	6.891	0.229
	10	12.268	0.268	9.005	0.532
	20	18.510	0.554	18.443	0.558
B1.3	5	8.481	0.132	2.992	0.701
	10	12.741	0.239	4.083	0.944
	20	19.289	0.503	13.987	0.831
NA1.1	5	8.317	0.140	4.130	0.531
	10	11.592	0.313	6.822	0.742
	20	17.655	0.610	17.389	0.628
NA1.2	5	8.842	0.116	4.148	0.528
	10	12.076	0.280	7.185	0.708
	20	18.323	0.566	17.705	0.607

G Main suite forecast results

Table 11: Out-of-sample forecast accuracy by model, scheme, and horizon. Robustness variants (R1.2–R1.4) were estimated under the rolling-window scheme only.

Model	Rolling window		Increasing window	
	$h = 1$	$h = 5$	$h = 1$	$h = 5$
<i>Panel A: QLIKE</i>				
B1.1	−7.655	−6.310	−7.661	−6.307
B1.2	−7.758	−6.367	−7.758	−6.365
B1.3	−7.767	−6.375	−7.773	−6.375
NA1.1	−7.655	−6.310	−7.661	−6.307
NA1.2	−7.641	−6.298	−7.638	−6.292
R1.2	−7.654	−6.308	–	–
R1.3	−7.639	−6.305	–	–
R1.4	−7.655	−6.310	–	–
<i>Panel B: RMSE</i>				
B1.1	0.0006	0.0018	0.0006	0.0018
B1.2	0.0006	0.0020	0.0006	0.0020
B1.3	0.0006	0.0016	0.0006	0.0016
NA1.1	0.0006	0.0018	0.0006	0.0018
NA1.2	0.0006	0.0018	0.0006	0.0018
R1.2	0.0006	0.0018	–	–
R1.3	0.0006	0.0018	–	–
R1.4	0.0006	0.0018	–	–

H DM Test results

Table 12: Diebold-Mariano tests of equal predictive accuracy. $\Delta_{QL} = \text{QLIKE}(\text{model}) - \text{QLIKE}(\text{baseline})$; negative values favour the candidate model. Newey-West standard errors with bandwidth $h - 1$ for $h > 1$.

Model	Baseline	Scheme	h	Δ_{QL}	DM	p	Result
B1.2	B1.1	RW	1	-0.1022	-2.131	0.033	Better
B1.2	B1.1	RW	5	-0.0571	-2.029	0.042	Better
B1.2	B1.1	IW	1	-0.0972	-2.095	0.036	Better
B1.2	B1.1	IW	5	-0.0581	-2.235	0.025	Better
B1.3	B1.1	RW	1	-0.1118	-1.722	0.085	No diff.
B1.3	B1.1	RW	5	-0.0654	-1.953	0.051	No diff.
B1.3	B1.1	IW	1	-0.1125	-1.814	0.070	No diff.
B1.3	B1.1	IW	5	-0.0682	-2.204	0.028	Better
NA1.1	B1.1	RW	1	0.0000	0.058	0.954	No diff.
NA1.1	B1.1	RW	5	-0.0004	-1.611	0.107	No diff.
NA1.1	B1.1	IW	1	-0.0004	-1.372	0.170	No diff.
NA1.1	B1.1	IW	5	0.0000	0.099	0.921	No diff.
NA1.2	B1.1	RW	1	0.0146	1.233	0.218	No diff.
NA1.2	B1.1	RW	5	0.0117	1.524	0.128	No diff.
NA1.2	B1.1	IW	1	0.0225	1.623	0.105	No diff.
NA1.2	B1.1	IW	5	0.0150	1.399	0.162	No diff.
R1.2	B1.1	RW	1	0.0012	0.889	0.374	No diff.
R1.2	B1.1	RW	5	0.0013	1.916	0.055	No diff.
R1.3	B1.1	RW	1	0.0168	1.231	0.218	No diff.
R1.3	B1.1	RW	5	0.0045	0.618	0.537	No diff.
R1.4	B1.1	RW	1	-0.0001	-0.215	0.830	No diff.
R1.4	B1.1	RW	5	-0.0002	-1.000	0.317	No diff.

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Table 12 continued

Model	Baseline	Scheme	h	Δ_{QL}	DM	p	Result
B1.1	B1.2	RW	1	0.1022	2.131	0.033	Baseline better
B1.1	B1.2	RW	5	0.0571	2.029	0.042	Baseline better
B1.1	B1.2	IW	1	0.0972	2.095	0.036	Baseline better
B1.1	B1.2	IW	5	0.0581	2.235	0.025	Baseline better
B1.3	B1.2	RW	1	-0.0096	-0.256	0.798	No diff.
B1.3	B1.2	RW	5	-0.0083	-0.689	0.491	No diff.
B1.3	B1.2	IW	1	-0.0154	-0.442	0.659	No diff.
B1.3	B1.2	IW	5	-0.0100	-0.904	0.366	No diff.
NA1.1	B1.2	RW	1	0.1022	2.135	0.033	Baseline better
NA1.1	B1.2	RW	5	0.0567	2.022	0.043	Baseline better
NA1.1	B1.2	IW	1	0.0968	2.094	0.036	Baseline better
NA1.1	B1.2	IW	5	0.0581	2.245	0.025	Baseline better
NA1.2	B1.2	RW	1	0.1168	2.238	0.025	Baseline better
NA1.2	B1.2	RW	5	0.0688	2.212	0.027	Baseline better
NA1.2	B1.2	IW	1	0.1196	2.318	0.021	Baseline better
NA1.2	B1.2	IW	5	0.0732	2.471	0.014	Baseline better
R1.2	B1.2	RW	1	0.1034	2.154	0.031	Baseline better
R1.2	B1.2	RW	5	0.0584	2.072	0.038	Baseline better
R1.3	B1.2	RW	1	0.1190	2.316	0.021	Baseline better
R1.3	B1.2	RW	5	0.0615	2.070	0.039	Baseline better
R1.4	B1.2	RW	1	0.1021	2.135	0.033	Baseline better
R1.4	B1.2	RW	5	0.0569	2.027	0.043	Baseline better

I Sub-Sample results

Table 13: Sub-sample OOS evaluation across regimes: 2019 (normal), 2020–21 (COVID), 2022–23 (inflation shock).

Sub-sample	Model	Scheme	h	T	QLIKE	RMSE
<i>2019 — normal</i>						
2019	B1.2	IW	1	50	-8.6880	0.0001
2019	B1.3	IW	1	50	-8.6670	0.0001
2019	NA1.2	IW	1	50	-8.5838	0.0001
2019	NA1.1	IW	1	50	-8.5673	0.0001
2019	B1.1	IW	1	50	-8.5673	0.0001
2019	B1.3	IW	5	50	-7.0319	0.0004
2019	B1.2	IW	5	50	-7.0274	0.0004
2019	NA1.2	IW	5	50	-6.9681	0.0004
2019	B1.1	IW	5	50	-6.9615	0.0004
2019	NA1.1	IW	5	50	-6.9612	0.0004
2019	B1.2	RW	1	50	-8.6881	0.0001
2019	B1.3	RW	1	50	-8.6581	0.0001
2019	NA1.2	RW	1	50	-8.5822	0.0001
2019	R1.3	RW	1	50	-8.5807	0.0001
2019	NA1.1	RW	1	50	-8.5637	0.0001
2019	B1.1	RW	1	50	-8.5630	0.0001
2019	R1.4	RW	1	50	-8.5629	0.0001
2019	R1.2	RW	1	50	-8.5565	0.0001
2019	B1.3	RW	5	50	-7.0289	0.0004
2019	B1.2	RW	5	50	-7.0270	0.0004
2019	NA1.2	RW	5	50	-6.9730	0.0003

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Table 13 continued

Sub-sample	Model	Scheme	h	T	QLIKE	RMSE
2019	R1.3	RW	5	50	-6.9679	0.0003
2019	NA1.1	RW	5	50	-6.9642	0.0004
2019	R1.4	RW	5	50	-6.9638	0.0004
2019	B1.1	RW	5	50	-6.9635	0.0004
2019	R1.2	RW	5	50	-6.9588	0.0004
2020-21 — COVID						
2020-21	B1.2	IW	1	101	-7.0658	0.0009
2020-21	B1.3	IW	1	101	-7.0646	0.0009
2020-21	NA1.1	IW	1	101	-6.9918	0.0009
2020-21	B1.1	IW	1	101	-6.9909	0.0009
2020-21	NA1.2	IW	1	101	-6.9368	0.0009
2020-21	B1.3	IW	5	101	-6.0925	0.0025
2020-21	B1.2	IW	5	101	-6.0830	0.0031
2020-21	NA1.1	IW	5	101	-5.9873	0.0027
2020-21	B1.1	IW	5	101	-5.9870	0.0027
2020-21	NA1.2	IW	5	101	-5.9694	0.0027
2020-21	B1.2	RW	1	101	-7.0635	0.0009
2020-21	B1.3	RW	1	101	-7.0497	0.0009
2020-21	R1.2	RW	1	101	-6.9756	0.0009
2020-21	R1.4	RW	1	101	-6.9750	0.0009
2020-21	NA1.1	RW	1	101	-6.9748	0.0009
2020-21	B1.1	RW	1	101	-6.9746	0.0009

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Table 13 continued

Sub-sample	Model	Scheme	h	T	QLIKE	RMSE
2020-21	NA1.2	RW	1	101	-6.9339	0.0009
2020-21	R1.3	RW	1	101	-6.9316	0.0009
2020-21	B1.3	RW	5	101	-6.0860	0.0025
2020-21	B1.2	RW	5	101	-6.0849	0.0031
2020-21	NA1.1	RW	5	101	-5.9903	0.0027
2020-21	R1.4	RW	5	101	-5.9900	0.0027
2020-21	B1.1	RW	5	101	-5.9896	0.0027
2020-21	R1.2	RW	5	101	-5.9889	0.0027
2020-21	R1.3	RW	5	101	-5.9781	0.0027
2020-21	NA1.2	RW	5	101	-5.9669	0.0027
2022-23 — inflation shock						
2022-23	B1.3	IW	1	101	-8.0399	0.0003
2022-23	B1.2	IW	1	101	-7.9899	0.0003
2022-23	B1.1	IW	1	101	-7.8822	0.0003
2022-23	NA1.1	IW	1	101	-7.8821	0.0003
2022-23	NA1.2	IW	1	101	-7.8721	0.0003
2022-23	B1.3	IW	5	101	-6.3332	0.0008
2022-23	B1.2	IW	5	101	-6.3199	0.0008
2022-23	B1.1	IW	5	101	-6.3035	0.0009
2022-23	NA1.1	IW	5	101	-6.3033	0.0009
2022-23	NA1.2	IW	5	101	-6.2803	0.0009
2022-23	B1.3	RW	1	101	-8.0436	0.0003

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Table 13 continued

Sub-sample	Model	Scheme	h	T	QLIKE	RMSE
2022-23	B1.2	RW	1	101	-7.9911	0.0003
2022-23	B1.1	RW	1	101	-7.8869	0.0003
2022-23	R1.4	RW	1	101	-7.8867	0.0003
2022-23	NA1.1	RW	1	101	-7.8863	0.0003
2022-23	R1.2	RW	1	101	-7.8860	0.0003
2022-23	NA1.2	RW	1	101	-7.8817	0.0003
2022-23	R1.3	RW	1	101	-7.8791	0.0003
2022-23	B1.3	RW	5	101	-6.3402	0.0008
2022-23	B1.2	RW	5	101	-6.3215	0.0008
2022-23	B1.1	RW	5	101	-6.3059	0.0009
2022-23	R1.4	RW	5	101	-6.3058	0.0009
2022-23	NA1.1	RW	5	101	-6.3057	0.0009
2022-23	R1.2	RW	5	101	-6.3056	0.0009
2022-23	R1.3	RW	5	101	-6.3040	0.0009
2022-23	NA1.2	RW	5	101	-6.2946	0.0009